Circuit Analysis of Tool Trust in Tool-Integrated Reasoning

Relevant Past Papers

- When to Trust Tools? Adaptive Tool Trust Calibration For Tool-Integrated Math Reasoning (https://arxiv.org/pdf/2604.08281)
    - Shows that open-source TIR models systematically distrust correct code-interpreter outputs when they conflict with the model's own reasoning ("Tool Ignored"), and proposes ATTC, a confidence-score-based framework that decides when to trust tool results, gaining 4.1–7.5% accuracy
    - *Gap:* doesn't explain where or how the trust decision is computed in the model
- Tool Calling is Linearly Readable and Steerable in Language Models (https://arxiv.org/pdf/2605.07990)
    - Across 12 models (270M–27B), shows tool *identity* is linearly decodable from the residual stream and intervenable, using probing + SAEs + steering
    - *Gap:* only covers tool *selection* (which tool to call)
- Internal Representations as Indicators of Hallucinations in Agent Tool Selection (https://arxiv.org/pdf/2601.05214)
    - Hidden-state probes detect hallucinated/incorrect tool calls in the same forward pass, before they are emitted
    - *Gap:* only covers tool selection
- Model-Adaptive Tool Necessity Reveals the Knowing-Doing Gap in LLM Tool Use (https://arxiv.org/pdf/2605.14038)
    - Redefines tool-necessity per model, finds 26.5–54% mismatch between needed vs actual tool calls, and probes hidden states to show a "knowing-doing gap": models internally know a tool is needed but fail to act on it, with the failure in the cognition-to-action transition
    - Gap: only covers the pre-execution decision to call a tool, never whether a returned tool output is trusted or overridden

Motivation

Tool-Integrated Reasoning (TIR) has become the dominant paradigm for equipping large reasoning models with capabilities they intrinsically lack, such as calculators and code interpreters. The value of this paradigm rests on the assumption that the model will actually use the returned results. Xu et al. (2026) disprove this assumption, demonstrating when a tool result conflicts with the model's prior chain-of-thought, the model tends to side with its own reasoning and ignore correct execution outputs.

Current literature has not yet provided sufficient explanations to *why* a given model mis-arbitrates or *where* in the network the decision is made. We propose to close this gap by localizing and causally characterizing the internal circuit that arbitrates between a returned tool result and the model's own chain-of-thought, providing the first mechanistic analysis of the result-assimilation stage in tool use.

Key Ideas/Contributions/Novelty

RQ: Where and how does a TIR model decide to accept or override a returned tool result, and can that mechanism be causally steered?

Contributions:

1. **Mechanistic account of tool trust:** We localize and causally validate the components that jointly determine the accept/override decision, and we characterize when in depth and at which token positions the decision becomes readable.
    a. Circuit localization/discovery
    b. Universality
2. **Training-free, token-free intervention**: A head-scaling policy and a steering direction, derived from the discovered circuit and applied only at post-output token positions, that reduce the Tool-Ignored rate at zero inference-token overhead
    a. Compare vs attc and other similar methods

Methods

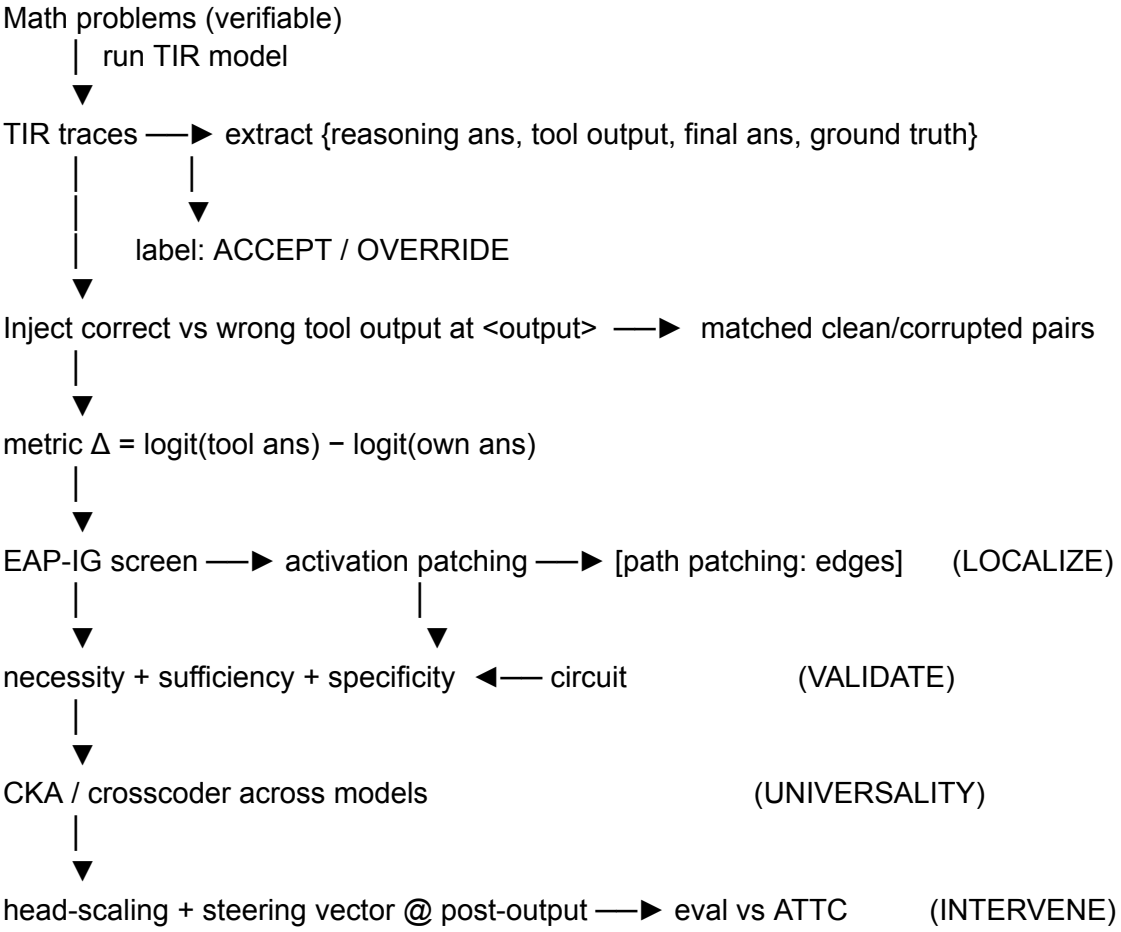
Our method proceeds from data construction, to a validated circuit, to a training-free steering fix, in the stages below.

1. **Tool-Conflict dataset.** Run TIR models on checker-verifiable math problems. From each trace extract four observable values (reasoning answer, tool output, final answer, ground truth) and label each as accept or override. Build matched clean/corrupted pairs by injecting a correct vs a wrong tool output at the `<output>` marker on the same problem (holds everything fixed except the tool value).
2. **Behavioral metric.** Define the logit difference `Δ = logit(tool answer) - logit(own answer)` at post-output positions. Δ measures how strongly the model leans toward the tool vs its own reasoning; the sign predicts accept vs override.
3. **Circuit localization.** Screen all heads/MLPs with EAP-IG (ranked by effect on Δ), confirm the top candidates with activation patching between matched pairs, then (stretch) recover edges with path patching.
4. **Circuit validation.** Test the discovered circuit for necessity (ablation vs random-head controls), sufficiency/faithfulness (circuit alone recovers most of Δ), and specificity (agreement traces and general math accuracy unchanged), with self-repair checks.

5. **Universality.** Repeat localization across TIR models; measure representational similarity with CKA (weak) and crosscoders/matched components (strong); report which holds.
6. **Intervention (Contribution 2).** From the circuit, derive a head-scaling policy and a difference-of-means "trust the tool" steering vector, applied only at post-output positions. Evaluate reduction in Tool-Ignored rate and accuracy gain vs ATTC at zero added tokens.

DLA + EAP-IG screen -> activation patching -> ablation/faithfulness validation -> CKA. Feasible, IOI-style, covers both contributions.

Diagram (pipeline):



Experimental Setup

How we test it: We construct conflict cases (matched accept vs override on the same problem), locate the components whose activation determines the outcome, then intervene on exactly those components and check the override rate drops. The logic mirrors a controlled experiment: the injected tool value is the manipulated variable, Δ and the final answer are the responses, and matched pairs hold all else constant.

Baselines (control groups).

- No-intervention base model (lower bound).
- Random-head scaling/steering (controls for "any intervention helps").
- ATTC (Xu et al. 2026), the existing behavioral fix, reproduced on our models.

Models: Open TIR math models with accessible activations: ToRL-7B, SimpleTIR-7B, VerlTool-7B (Qwen2.5-Math-7B base); loaded in TransformerLens/nnsight. Multiple models support the universality claim.

Datasets: MATH500, GSM8K, Minerva, AIME24, AMC23 (same as Xu et al., for comparability).

Metrics: Logit difference Δ; Tool-Ignored rate; end-task accuracy; circuit faithfulness (% of Δ recovered); CKA.

Additional analysis / confound control: Random-head controls isolate our circuit from arbitrary heads; specificity tests confirm interventions leave accept-cases and general math accuracy intact; self-repair checks (iterated ablation) guard against backup-head masking; results repeated across corruption types and seeds. Visualizations: per-layer/per-position Δ attribution heatmap, before/after intervention rates.

Datasets and Evaluation

Xu et al. (2026) show reasoning-vs-tool conflicts are frequent (40–60% of wrong-answer cases) and Tool-Ignored is ≥15% of errors; their released traces and conflict-detection script provide natural accept and override cases. We use two layers:

- **Natural set (from their data):** labeled accept/override traces for realism, rate measurement.
- **Injected set (ours):** matched clean/corrupted pairs (correct vs wrong tool output on the same problem) for the controlled contrast that patching, and steering require.

We do not train the model. Data is split into a discovery set (circuit finding) and a held-out set (intervention evaluation). Evaluation is checker-verifiable (auto-graded math), plus Tool-Ignored rate.

Benchmarks/Evaluation Sets

- Same math benchmarks (MATH500, GSM8K, Minerva, AIME24, AMC23) on held-out data. Baselines: no-intervention base, random-head control, and ATTC. The key comparison: match ATTC's accuracy gain (≥4%) and Tool-Ignored reduction at zero added tokens and no training.

Ideal Results

Hypothesis: a small, localized set of components at post-output positions causally arbitrates accept vs override, recurs across models, and can be steered.

- **Localization:** <2% of heads (mid-to-late layers, post-output positions) mediate the decision; patching them recovers ≥70% of Δ.
- **Universality:** the circuit recurs across ToRL/SimpleTIR/VerlTool (high CKA at the relevant layers).
- **Intervention:** steering/head-scaling cuts Tool-Ignored rate ≥50% relative and matches ATTC's accuracy gain (≥4%) with no training and zero added tokens, without inflating over-trust on wrong-tool cases.

Potential Limitations

- **Compute:** full edge-level circuits (path patching, attribution graphs) over long traces are expensive; the backbone is head/MLP-level localization, with edges as a stretch.
- **Generalization:** circuits found on 7B models may not transfer to larger or closed models.
- **Dataset:** injected conflicts are synthetic; we validate against natural conflict cases.
- **Method:** self-repair can mask necessity; steering may trade Tool-Ignored reduction against over-trust on genuinely wrong tool outputs.
- **Ethical:** steering "trust the tool" could be misused to force acceptance of faulty outputs; the fix assumes tool correctness is verifiable.